Answer **all** the questions in this section in the spaces provided.

1 A solar propulsion engine uses solar power to ionize and accelerate atoms of xenon. The speed of the ejected xenon ions relative to the spaceship is 3.0×10^4 m s^{-1} as shown in Fig. 1.1.

Fig. 1.1

Fig. 1.2 shows the variation with time *t* of the acceleration *a* of the spaceship as a result of the ejection of xenon ions.

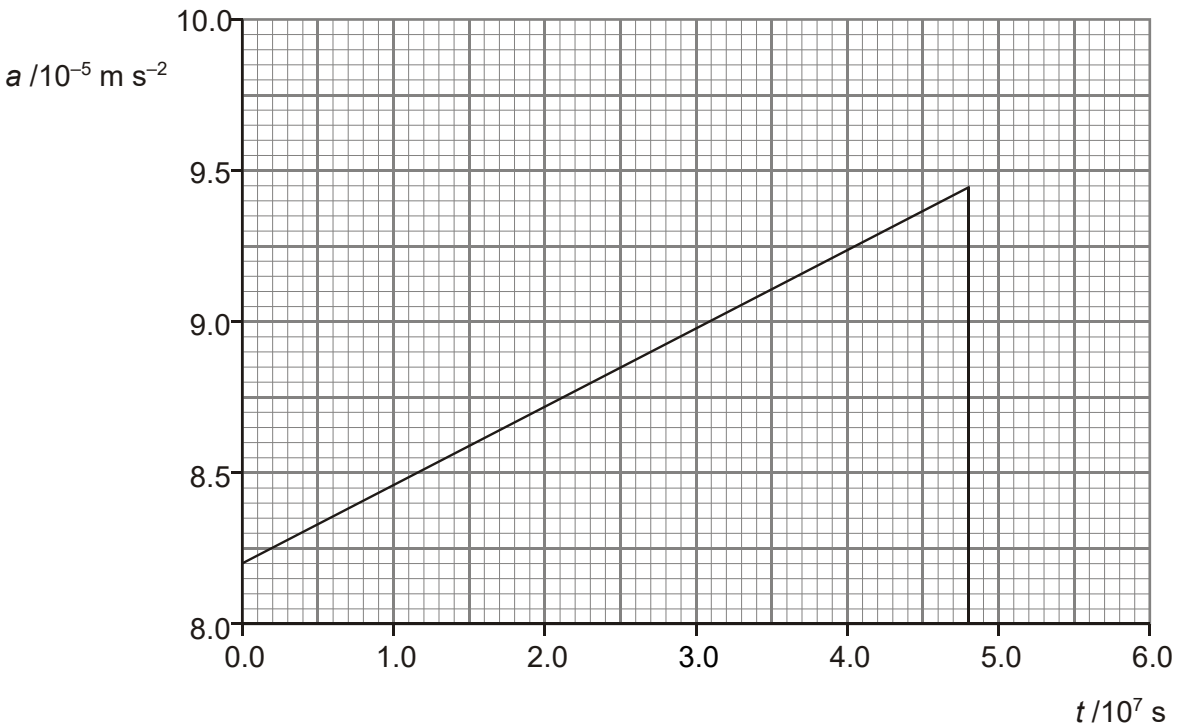

Fig. 1.2

(a) The xenon ions are ejected at a constant rate of 1.7×10^{-6} kg s^{-1}. Calculate the force exerted on the spaceship by the xenon ions.

force = ... N [2]

(b) Explain why the acceleration of the spaceship is increasing with time.

.....

.....

..... [1]

(c) The solar propulsion engine is switched on at time t = 0 when the initial velocity of the spaceship is zero.

Use Fig. 1.2 to determine the final velocity of the spaceship when the fuel runs out.

velocity = ... m s^{-1} [3]

(d) Sketch on Fig. 1.3 the corresponding variation with time t of the velocity v of the spaceship from t = 0 to t = 6.0 x 10^7 s.

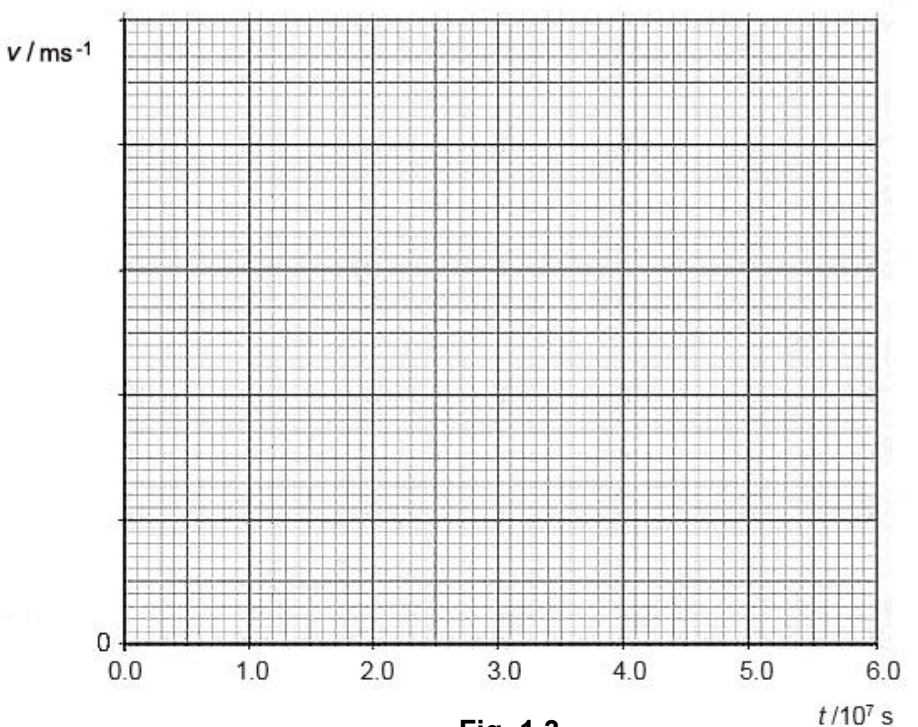

Fig. 1.3

[2]